

BASICS FOR AIR TRAFFIC CONTROL – SEARCH AND RESCUE (SAR)

INTRODUCTION

There are a large number of civil aircraft that fly our airspace. Each aircraft is operated by pilots with various levels of flying experience. Occasionally, one of them flies off course, is overdue, or experiences a mechanical or medical emergency. Depending on the situation, a search and rescue mission may be required. The Federal Aviation Administration (FAA), along with a cooperative effort of various agencies, plays a part in assisting pilots in need of help.

The purpose of this module is to cover Search and Rescue (SAR) responsibilities, handling, and procedures. It also covers the actions taken when an aircraft becomes unreported, overdue, or missing.



SEARCH AND RESCUE (SAR) RESPONSIBILITIES

Purpose: This lesson covers responsibilities to conduct effective civil search and rescue missions.

Objective:

- Identify Search and Rescue (SAR) responsibilities to conduct effective civil SAR missions

References for this lesson are as follows:

- FAA Order JO 7110.10, Flight Services
- FAA Order JO 7110.65, Air Traffic Control
- Aeronautical Information Manual (AIM)
- National Search and Rescue Plan

Purpose of National Search and Rescue Plan

National Search and Rescue Plan – An interagency agreement which provides for the effective utilization of all available resources in all types of civil SAR missions.

Participants include but not limited to:



Federal Aviation
Administration (FAA)



United States Coast
Guard (USCG)



U.S. AIR FORCE

United States Air
Forces (USAF)



National Aeronautics
and Space
Administration (NASA)

Search and Rescue – A service which seeks missing aircraft and assists those found to be in need of assistance. It is a cooperative effort using the facilities and services of available Federal, state, and local agencies.

Note: The U.S. Coast Guard is responsible for coordination of search and rescue for the Maritime Region, and the U.S. Air Force is responsible for search and rescue for the Inland Region. Information pertinent to search and rescue should be passed through any air traffic facility or be transmitted directly to the Rescue Coordination Center by telephone.

Overdue

Overdue describes an aircraft that has failed to report and the appropriate time frame has elapsed (30 minutes or 1 hour).

Unreported

Unreported refers to the time between when an aircraft should have reported a compulsory reporting point until it is considered overdue.

Missing

Aircraft are considered **missing** after the communications search has failed to locate the aircraft.



Search and Rescue Facility

A **search and rescue facility** is responsible for **maintaining and operating a SAR service** to render aid to persons and property in distress.

It is any SAR unit, station, NET, or other operational activity which can be usefully employed during an SAR Mission; e.g., a Civil Air Patrol Wing, or a Coast Guard Station.

Examples include but are not limited to:

- Flight Service Stations (FSSs)
- Civil Air Patrol
- Rescue Coordination Centers (RCCs)
- USCG
- Local Law Enforcement



Knowledge Check A

REVIEW what you have learned so far about SAR responsibilities. ANSWER the questions listed below.

1. A provision for the effective utilization of all available facilities for SAR missions is called a/an _____.
(Select the correct answer.)
 - ☐ Search and Rescue Facility
 - ☒ National Search and Rescue Plan
 - ☐ Rescue Coordination Center
2. An aircraft that has failed to report and the appropriate time frame has elapsed describes a/an _____.
(Select the correct answer.)
 - ☐ Unreported
 - ☒ Overdue
 - ☐ Missing

Roles and Responsibilities: Federal Aviation Administration

Under the National Search and Rescue Plan, the Federal Aviation Administration (FAA):

- Provides emergency service(s) to aircraft in distress
- Ensures that SAR procedures are initiated if an aircraft is overdue or missing
- Attempts to locate aircraft through communications search via the:
 - ATC system for Instrument Flight Rules (IFR) aircraft
 - Flight plan program for Visual Flight Rules (VFR) aircraft

FAA SAR Facility Responsibility

TYPE OF FACILITY	FACILITY RESPONSIBILITIES
FSS	Flight Service Station (FSS) initiates SAR procedures for overdue or missing aircraft: ■ On a VFR flight plan ■ Where no flight plan has been filed ■ On the VFR portion of a composite VFR/IFR flight plan Note: Composite flight plans are used primarily by the military.
ARTCC	The Air Route Traffic Control Center (ARTCC) initiates SAR procedures for aircraft that are: ■ Overdue or missing on an IFR flight plan ■ On a combination VFR/IFR or an airfield IFR flight plan and 30 minutes have passed since the pilot requested IFR clearance, and neither communication nor radar contact can be established ■ On a Special VFR clearance ■ When an Emergency Locator Transmitter (ELT) signal is received or reported • ELT testing is done during the first five minutes of the hour. If testing is done outside this time frame, coordination should be accomplished with the nearest FAA Control Tower or FSS

Military Responsibility

The military coordinates and conducts physical SAR operations through RCCs.

- The U.S. Air Force is responsible for SAR for the Inland Region
- The USCG is responsible for coordination of SAR for the Maritime Region



Rescue Coordination Centers (RCCs)

Air Force RCCs

- Tyndall Air Force Base (AFB), FL, is responsible for SAR coordination for the inland region within the contiguous states
- Joint Base Elmendorf-Richardson is responsible for Alaska
- Usually uses civilian resources to conduct physical search, such as Civil Air Patrol and state agencies

Coast Guard (USCG) RCCs

Using various stations around the country, they are responsible for coordination and conducting SAR for the maritime region.

Joint RCCs

Hawaii Joint RCC, located in Honolulu, is responsible for Hawaii.

Inland & Maritime?





Knowledge Check B

REVIEW what you have learned so far about SAR roles and responsibilities and RCCs. ANSWER the questions listed below.

1. Which facility has initial SAR responsibility for aircraft on a VFR flight plan? *(Select the correct answer.)*
 - ☐ ARTCC
 - ☐ RCC
 - ☒ FSS
2. Which facility has SAR responsibility for aircraft on a Special VFR clearance? *(Select the correct answer.)*
 - ☒ ARTCC
 - ☐ RCC
 - ☐ FSS
3. Which AFB is responsible for SAR coordination for the inland region of the contiguous states? *(Select the correct answer.)*
 - ☐ Joint Base Elmendorf-Richardson
 - ☐ Scott AFB
 - ☒ Tyndall AFB

Search and Rescue Responsibilities Summary

The National Search and Rescue Plan interagency agreement provides for the effective utilization of all available resources in all types of civil search and rescue missions. This lesson also identified responsibilities which are necessary to conduct an effective search and rescue mission. This service, which seeks missing and assists those in need of assistance, is a cooperative effort for various agencies and is a vital task for the FAA.

OVERDUE AIRCRAFT PROCEDURES

Purpose: This lesson covers procedures for reporting unreported, overdue, or missing aircraft.

Objective:

- Identify procedures for reporting unreported, overdue, or missing aircraft

References for this lesson are as follows:

- FAA Order JO 7110.10, Flight Services
- FAA Order JO 7110.65, Air Traffic Control
- Aeronautical Information Manual (AIM)

Initiating Search and Rescue Service

The initiation of search and rescue (SAR) depends on whether or not a flight plan was filed and activated and the type of flight plan filed. Select each type of flight plan to learn when an aircraft is considered overdue and what your responsibilities are.

- Visual Flight Rules (VFR) flight plans
- Instrument Flight Rules (IFR) flight plans
- No flight plan

Visual Flight Rules (VFR) Flight Plans

An aircraft on a VFR or Defense Visual Flight Rules (DVFR) flight plan is considered overdue **30 minutes after the estimated time of arrival (ETA)** and communications or location cannot be established.

- $OVERDUE = ETA + 30 \text{ Minutes}$
- A communications search is initiated by a Flight Service Station (FSS)

Instrument Flight Rules (IFR) Flight Plans

An aircraft on an IFR flight plan is considered overdue when neither communications nor radar contact can be established and **30 minutes** have passed since its:

- ETA over a specified or compulsory reporting point or at a clearance limit
- Clearance void time *← Some clearances are given w/ an expiration time*

Ex. Plan disappears from radar
If you have reason to believe that the aircraft is overdue prior to 30 minutes, **take immediate action as required**. If an IFR aircraft is determined to be overdue, unreported, or missing, the SAR process starts at this stage.

Note: The center in whose area the aircraft is first overdue or unreported will make these determinations and is responsible for taking any subsequent action required.

No Flight Plan

An aircraft with no flight plan is considered overdue at the actual time a reliable source reports it to be **at least one hour late at the destination**.

- Reliable source may be family member, business associate, fixed base operator, etc.
- A communications search is initiated by an FSS

Search and Rescue Terminology

Information Request (INREQ) is a request originated by FSS for information concerning an overdue VFR aircraft.

Alert Notice (ALNOT) is a request originated by an FSS or ARTCC for an extensive communications search for overdue, unreported, or missing aircraft.



Information to be Forwarded to ARTCCs by Terminal Facilities

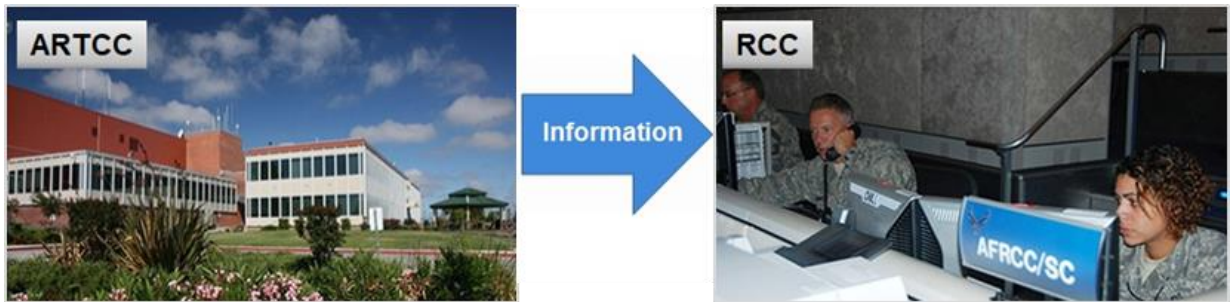
When an aircraft is considered to be in emergency status that may require SAR procedures or an IFR aircraft is overdue, the terminal facility shall alert the ARTCC and forward information, as available. Select the arrow to see the information that is required.



- Flight plan, including color or aircraft, if known
- Time of last transmission received, by whom, and frequency used
- Last position report and how determined
- Aircraft beacon code
- Number of persons on board
- Fuel status
- Facility working the aircraft and frequency
- Last known position, how determined, time, estimated present position, and maximum range of flight of the aircraft based on remaining fuel and airspeed
- Position of other aircraft near the aircraft's route of flight, when requested
- Whether or not an ELT signal has been heard or reported in the vicinity of the last known position
- Other pertinent information

Information to be Forwarded to RCC by ARTCCs

When an aircraft is considered to be in emergency status or an IFR aircraft is overdue, the ARTCC must alert RCC and forward the following information, as available. Select the arrow to see the information that is required.



- + ■ Facility and person calling
 - Flight plan, including color of aircraft, if known
 - Time of last transmission received, by whom, and frequency used
 - Last position report and how determined
 - Aircraft beacon code
- + ■ Action taken by reporting facility and proposed action
 - Number of persons on board
 - Fuel status
 - Facility working the aircraft and frequency
 - Last known position, how determined, time, estimated present position, and maximum range of flight of the aircraft based on remaining fuel and airspeed
 - Position of other aircraft near the aircraft's route of flight, when requested
 - Whether or not an ELT signal has been heard or reported in the vicinity of the last known position
 - Other pertinent information

Issuance of Alert Notice (ALNOT)

In addition to routing to the Regional Operations Center (ROC) for the area in which the facility is located, issue an ALNOT to all centers and all concerned facilities generally 50 miles on either side of the route of flight from the last reported position to destination.

- Include the original or amended flight plan, as appropriate, and the last known position of the aircraft
- The search area may be expanded to the maximum range of the aircraft
- All information about the overdue aircraft is included in the ALNOT to assist in search activities

Note: An ALNOT must be issued before the RCC can begin SAR operations.

- Upon receipt of an INREQ or ALNOT, check the position records to determine whether the aircraft has contacted your facility
 - Notify the originator of the results or status of this check within one hour of the time the alert was received
 - Retain the alert in an active status, and immediately notify the originator of subsequent contact, until cancellation is received



ALNOT Search Area

Include pertinent information in the ALNOT that will aid the RCC and SAR teams in conducting the SAR mission. When known, include:

Last known position	
Time	
Aircraft beacon code	

Responsibility Transfer to RCC

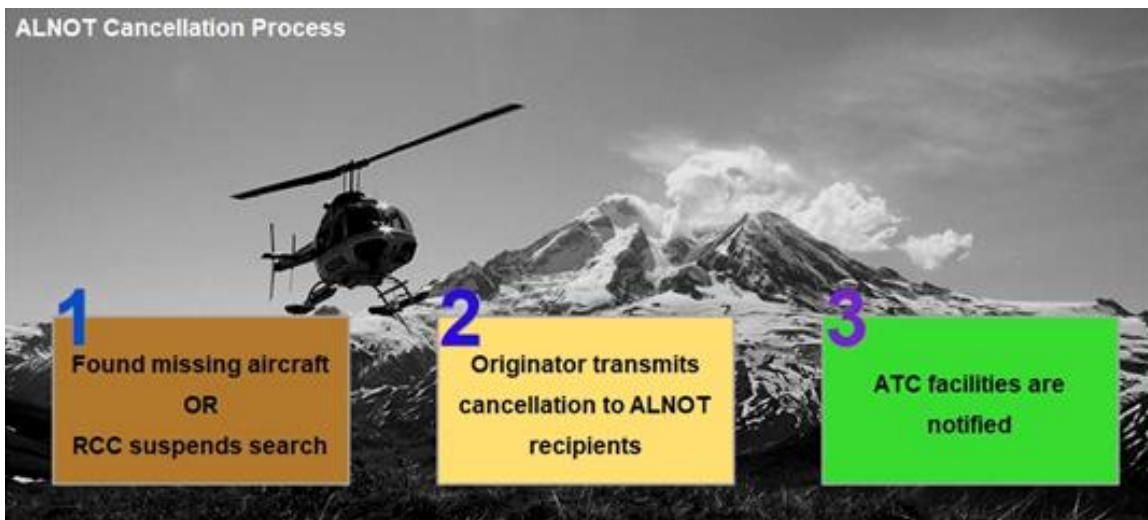
Transfer of search responsibility to RCC occurs when:

The ALNOT search has been completed with negative results	
The aircraft has not been located within one hour after ALNOT issuance	
Thirty minutes have elapsed after the estimated aircraft fuel exhaustion time	

Cancellation of ALNOT

The ALNOT remains active until the aircraft is located or RCC suspends the search.

- The ALNOT originator then transmits a cancellation message with the location of the aircraft, if found, to all recipients of the ALNOT
- ATC facilities are notified





Knowledge Check C

REVIEW what you have learned so far about conducting SAR procedures for overdue aircraft. ANSWER the questions listed below.

1. When is an aircraft on an IFR flight plan considered overdue? *(Select the correct answer.)*
 - ☐ Communication cannot be established over a mandatory reporting point
 - ☐ Radar contact cannot be established within 30 minutes of an aircraft's ETA
 - ☒ 30 minutes have passed since ETA over a reporting point or clearance limit and neither communication nor radar contact can be established
 - ☐ Not until FSS initiates a SAR
2. When is an aircraft on a VFR flight plan considered overdue? *(Select the correct answer.)*
 - ☐ At the actual time a reliable source reports it to be at least one hour late
 - ☐ Communication or radar cannot be established
 - ☒ 30 minutes after the ETA and communication or location cannot be established
 - ☐ 30 minutes have passed since ETA over a reporting point
3. When an aircraft is in emergency status and may require SAR procedures, what information must you forward to the ARTCC? *(Select all correct answers that apply.)*
 - ☒ Last position report and how determined
 - ☒ Number of persons on board
 - ☐ Information from last transmission received
 - ☒ Aircraft beacon code
 - ☒ Fuel Status
4. When an aircraft is considered to be in emergency status, which of the following information is the ARTCC **NOT** required to send to an RCC? *(Select the correct answer.)*
 - ☐ Flight plan, including color of aircraft, if known
 - ☐ Time of last transmission received, by whom, and frequency used
 - ☐ Number of persons on board
 - ☒ Next position report
 - ☐ Aircraft beacon code

Overdue Aircraft Procedures Summary

It is important to understand the point at which action must be taken to initiate a search and rescue. This lesson identified procedures for reporting unreported, overdue, or missing aircraft. This knowledge is vital to the FAA regarding the safety and livelihood of the aircraft in your care.

EXERCISE: IFR SEARCH AND RESCUE MATRIX

Purpose

This exercise is to help you become familiar with the search and rescue (SAR) process for aircraft on an IFR flight plan.

Directions

Read the **SAR Narrative**. Using the Search Matrix, answer the questions below.

Search Matrix	
PHASE OF SEARCH	AREA SEARCHED
INREQ	All airports along the route of flight that can accommodate the aircraft
ALNOT	The area extending 50 miles on either side of the route of flight from the last reported position to the destination
Question	Answer
1. At what time should an ALNOT message have been sent in reference to N2811H?	<i>Immediately</i>
2. To whom should the ALNOT message have be sent?	
3. What action should the stations receiving the ALNOT message have taken?	
4. What was the role of the RCC?	
5. Who was responsible for canceling the ALNOT?	

SAR Narrative

An advancing cold front had created a large area of clouds across the eastern plains of New Mexico and extreme western Texas. Many locations were reporting IFR ceilings. The forecast for the area showed little improvement until the slow-moving front passed by.

The pilot of N2811H, a Piper Cherokee, had to make a flight; an important business meeting in Albuquerque had been scheduled at the last minute. The pilot felt that flying in a rental would be easier than trying to get a ticket on an airline. Licensed and current in IFR operations, the pilot was confident in his abilities for a long cross-country flight. The flight across New Mexico would take N2811H along V402 to Tucumcari (TCC), then V12 to Albuquerque (ABQ).

Shortly after N2811H turned west onto V12, the Albuquerque Center (ZAB) sector controller noticed that the data block for N2811H was blinking erratically. Then it disappeared completely. Attempts to contact the aircraft by radio were unsuccessful. The sector controller notified the supervisor. The supervisor then advised the operations manager in charge of operations. An emergency was declared, and an ALNOT was issued to notify all affected Air Traffic facilities of the disappearance of N2811H.

The ALNOT started an immediate communication search of all airports and Air Traffic facilities between TCC and ABQ. ABQ FSS made many phone calls to all the small airports scattered across the eastern plains of New Mexico. The RCC alerted the Civil Air Patrol (CAP), who then alerted pilots and observers for air searches. Cannon AFB was also alerted. The State Highway Patrol was notified, as were county sheriff's offices along the route. A vast army of people had been formed in a short period, all looking for N2811H. Traffic files and flight lines were checked. Deputies and troopers kept their eyes open while patrolling their areas. Controllers in different facilities listened for that important call sign to call.

Approximately 35 minutes after the ALNOT went out, ABQ FSS received a phone call on its 800 briefing line. The pilot of N2811H reported that he was safely on the ground at Sandia Airpark (NM51). An electrical failure had shut down the instruments, radios, and transponder. Flying above the clouds, the pilot used the magnetic compass, charts, and a cool head to continue west. Breaks in the clouds revealed an airport. The pilot carefully descended through the break and landed safely.

ABQ FSS immediately sent this information to ZAB. With the aircraft located, ZAB canceled the ALNOT. The word was spread among those searching, and people went back to their normal routines.

SUMMARY

This purpose of this module was to cover Search and Rescue (SAR) responsibilities, handling, and procedures. It also covered the actions taken when an aircraft becomes unreported, overdue, or missing.

In accordance with FAA Order JO 7110.10, Flight Services; FAA Order JO 7110.65, Air Traffic Control; Aeronautical Information Manual (AIM); and National Search and Rescue Plan; you should now be able to:

- Identify Search and Rescue (SAR) responsibilities to conduct effective civil SAR missions
- Identify procedures for reporting unreported, overdue, or missing aircraft

KNOWLEDGE CHECK ANSWERS

Knowledge Check A

1. National Search and Rescue Plan; 2. Overdue

Knowledge Check B

1. FSS; 2. ARTCC; 3. Tyndall AFB

Knowledge Check C

1. 30 minutes have passed since ETA over a reporting point or clearance limit and neither communication nor radar contact can be established; 2. 30 minutes after the ETA and communication or location cannot be established; 3. Last position report and how determined, Number of persons on board, Aircraft beacon code, Fuel status; 4. Next position report